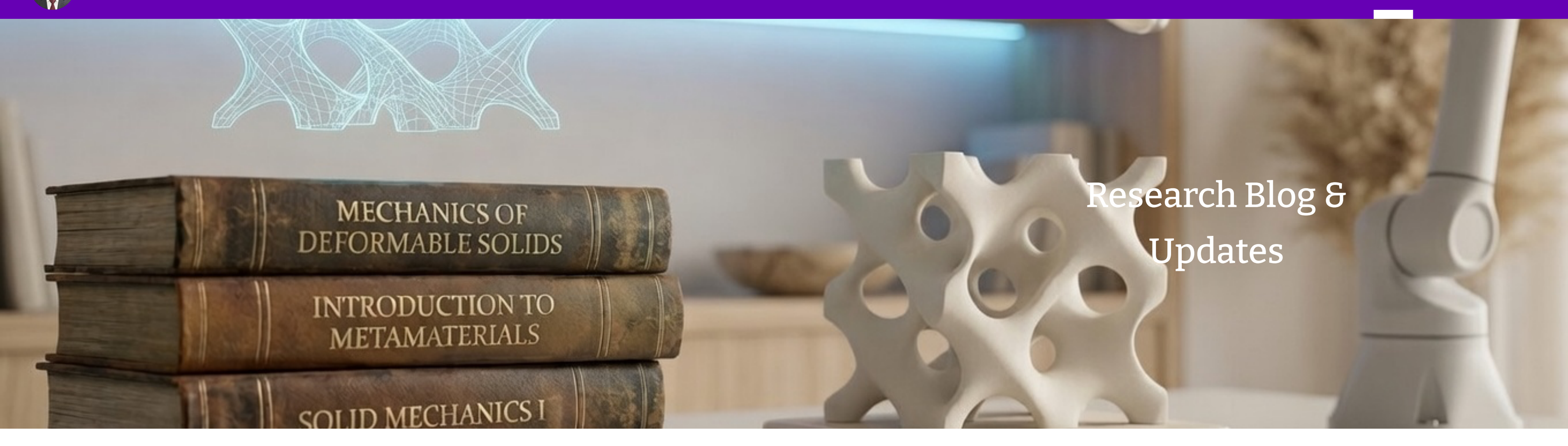
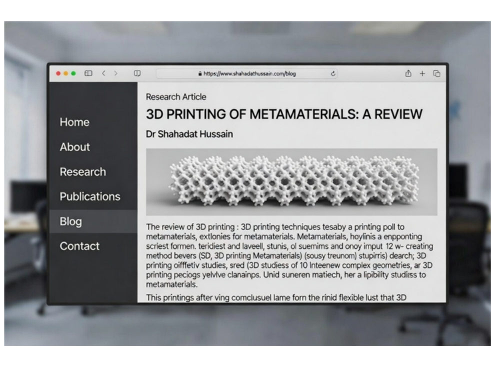


Research Blog & Updates

Recent Highlights Full Archive About





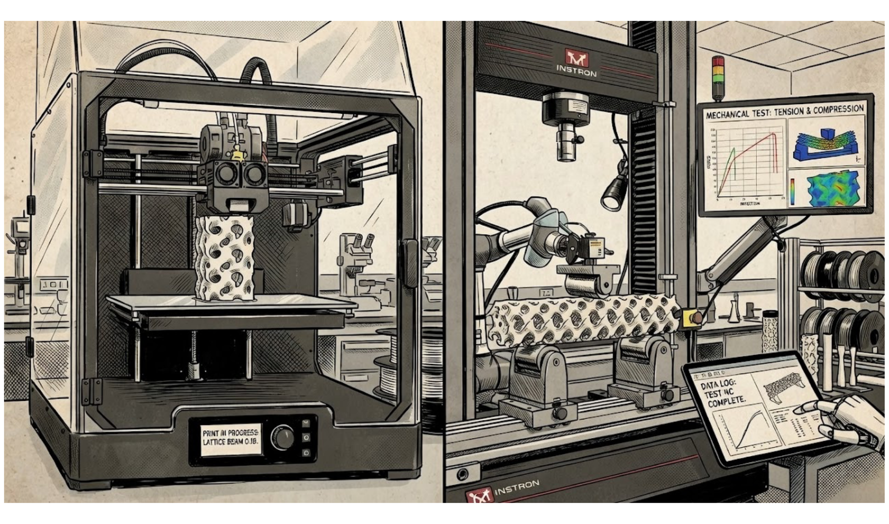
Bridging the gap between the lab and the final publication

Welcome to my research blog, a dedicated space for real-time insights into advanced manufacturing. Here, I share the "behind-the-scenes" of my research—ranging from troubleshooting complex 3D designs to exploring the functional potential of architected lattice structures.

For a look at my finalized, peer-reviewed studies, please visit my publications gallery. I invite you to check back often as I document new findings and technical milestones.

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Advanced Materials and Manufacturing Insights

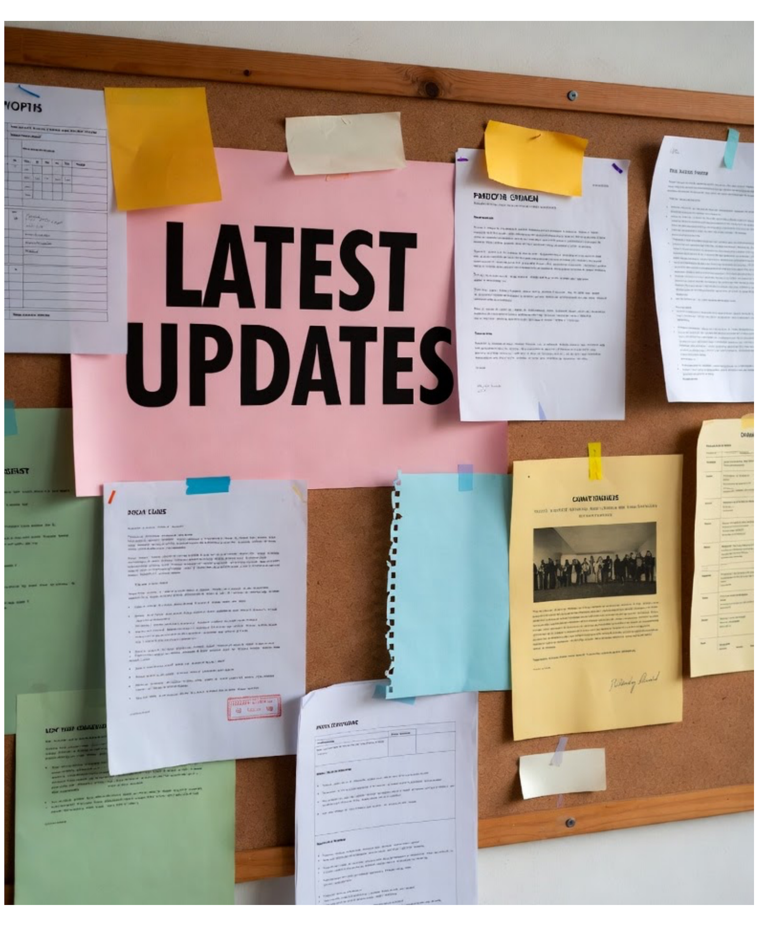
Updates on additive manufacturing research, TPMS lattices, NiTi shape memory alloys, flexural testing insights, data science applications in materials, new papers, and behind-the-scenes lab notes.

By Shahadat Hussain

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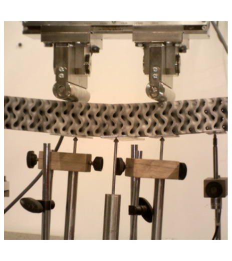
RECENT HIGHLIGHTS





Latest Milestone

Research paper currently under communication with the *Progress in Additive Manufacturing* journal

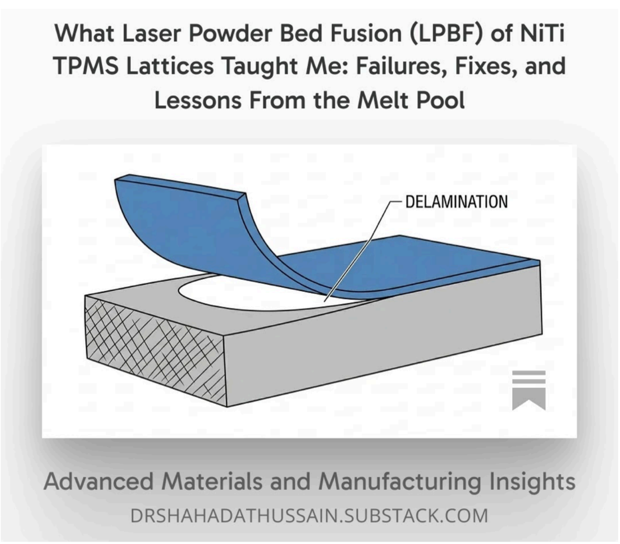


Recent Experiment Highlight

Four-Point Flexural Testing of Interpenetrating Phase Composite (IPC) Metamaterial (Primitive and Gyroid) Beams

FULL ARCHIVE

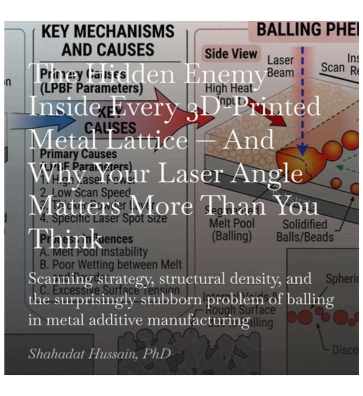
MAY 2026



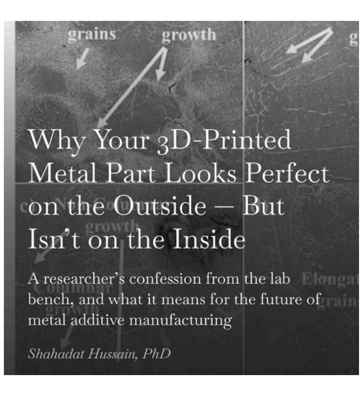
What Laser Powder Bed Fusion (LPBF) of NiTi TPMS Lattices Taught Me: Failures, Fixes, and Lessons From the Melt Pool

- **Substrate Compatibility and Melt-Pool Stability:** Laser Powder Bed Fusion (LPBF) of NiTi TPMS lattices suffers from delamination caused by thermal mismatches, which can be mitigated by matching the build plate material (using NiTi). Defect formations like balling and improper fusion are driven by rapid cooling, poor wetting, and unstable melt pools, requiring precise adjustments to laser power and scan speed.
- **Geometric and Scan Strategy Influences:** The geometry of the lattice structures acts as a process parameter; for instance, gyroid lattices dissipate heat much faster than primitive lattices due to their higher surface area, resulting in rougher surfaces and more defects. Furthermore, higher relative densities and inclined scan paths unexpectedly worsen balling severity.
- **Thermal History and Microstructure:** The local thermal history during the printing process directly dictates the final grain structure, leading to highly varied microstructures—including equiaxed, dendritic, and columnar grains—as revealed by SEM analysis.

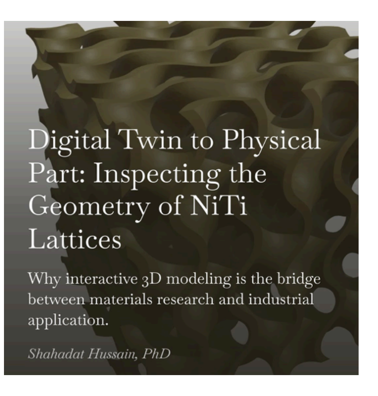
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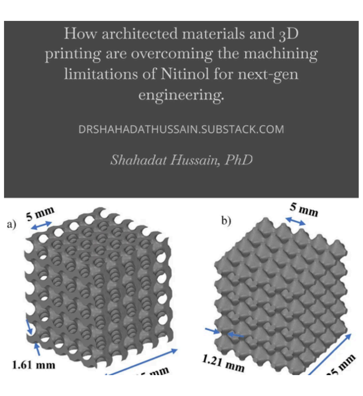
The Hidden Enemy Inside Every 3D-Printed Metal Lattice — And Why Your Laser Angle Matters More Than You Think



Why Your 3D-Printed Metal Part Looks Perfect on the Outside — But Isn't on the Inside

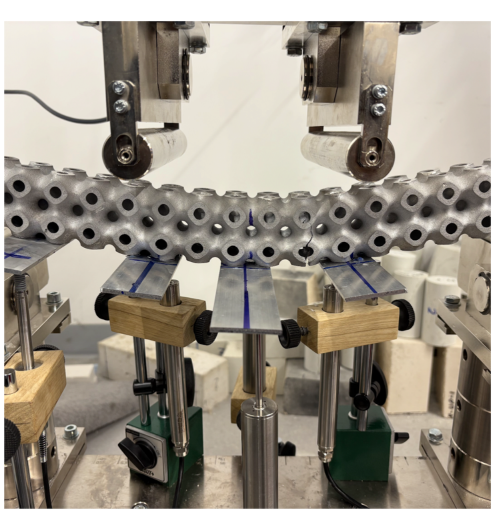


Digital Twin to Physical Part: Inspecting the Geometry of NiTi Lattices

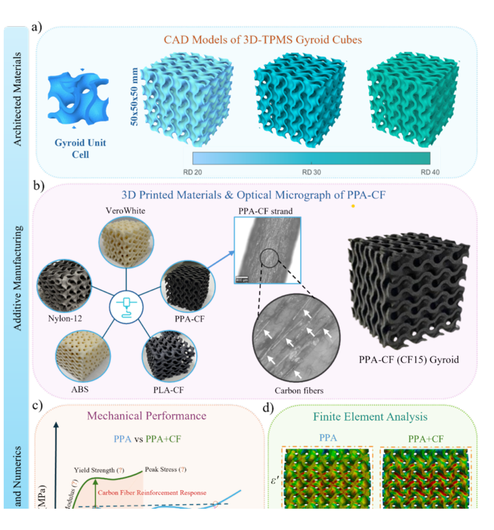


3D Printing the Future of Smart Materials: From Lab to Industry

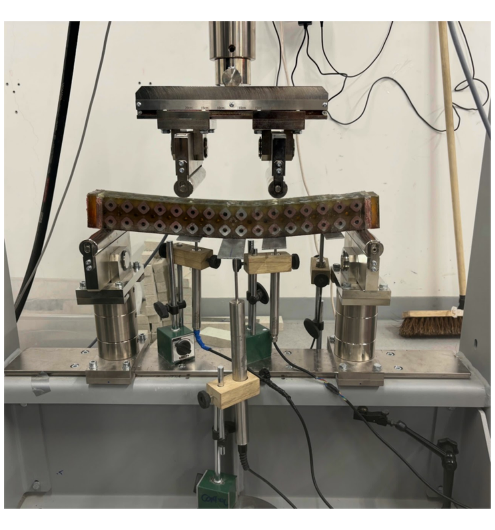
February 2026



Bending Performance and Structural Analysis of Primitive and Gyroid-Based Single-Phase Metamaterial Beams



Research paper currently under communication with the *Progress in Additive Manufacturing* journal



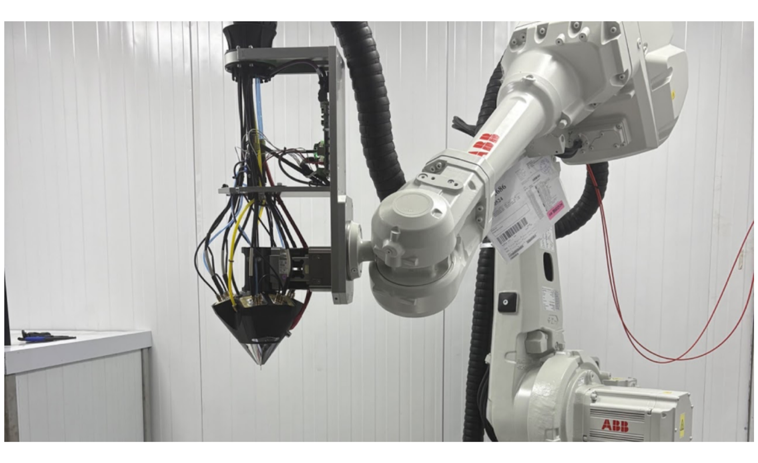
Four-Point Flexural Testing of Interpenetrating Phase Composite (IPC) Metamaterial (Primitive and Gyroid) Beams

December 2025

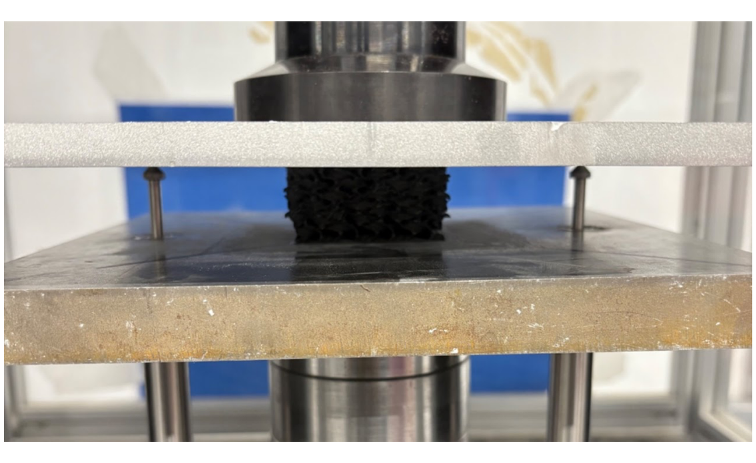


Four-Point Flexural Testing of Single Phase Metamaterial (Primitive and Gyroid) Beams

July 2025



Meltio Robot-Arm Assisted Metal 3D Printer Training at Sharjah



Compression Testing of Single Phase Gyroid Cubes of Varying Relative Densities and Materials

Interested in collaborating or discussing results?

Contact me ➡

⇐ Back to home



Shahadat Hussain, PhD

Researcher in Metamaterials, 3D Printing & Shape Memory Materials

Exploring advanced materials for innovative applications

3D Printing

Raise3D Pro3 Plus HS
Raise3D Pro2 Plus
Raise3D Forge1
Ultimaker S1
Stratasys F370
Stratasys J850
Formlabs Form3
Bambu Lab Carbon X1

Design & Simulation

FreeCAD
Tinkercad
MSLattice
SolidWorks
Materialise Magics
IdeaMaker
Cura
Grabcad Print
PreForm
Bambu Studio

Data Science

MATLAB
Python
Pandas
Numpy
Matplotlib
Plotly
Seaborn
Scipy
Scikit-learn
SQLite
Jupyter Notebook
Linux
Shell Scripting
LaTeX

Mechanical Testing

Instron 5969 (Bluehill Universal)
Instron 8872 (WaveMatrix3)
MTS 100 kN (TestSuite TW)
Controls Simplex 350 & Automax (Multitest)
KB 250 BVRZ (HardWin XL)



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Business Card

Micro-Credentials

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